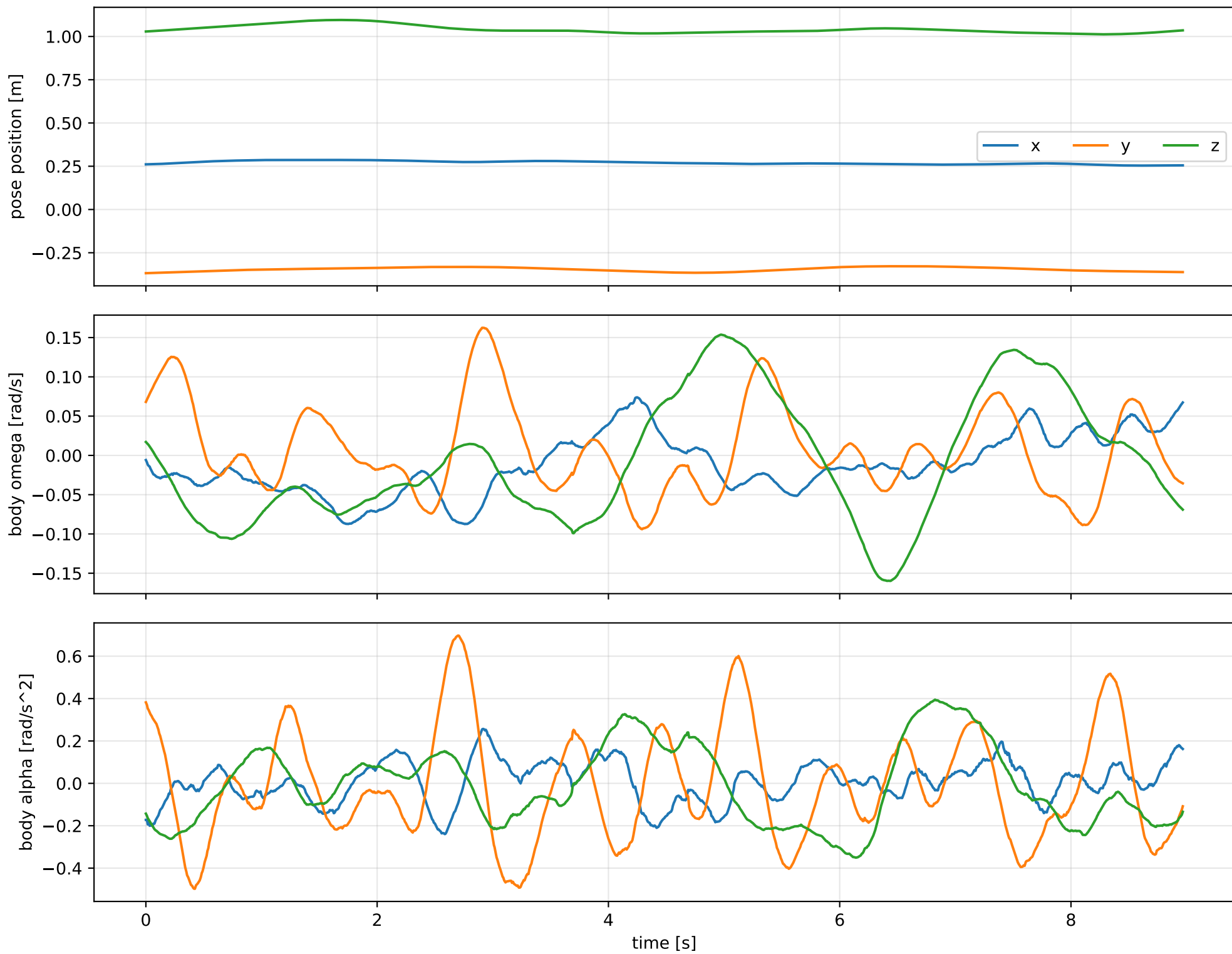
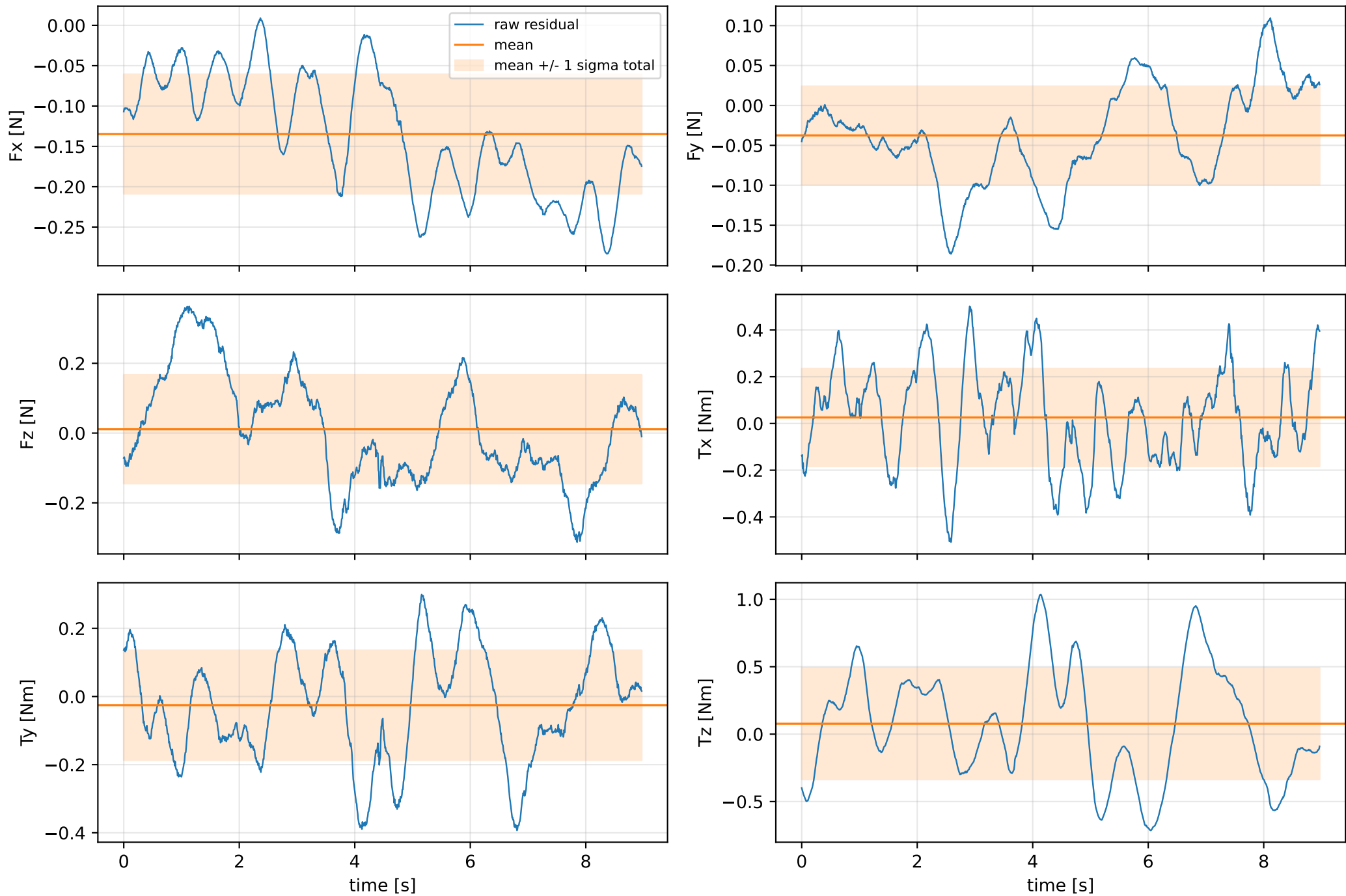


cog_y_nominal: SG trajectory and derived motion

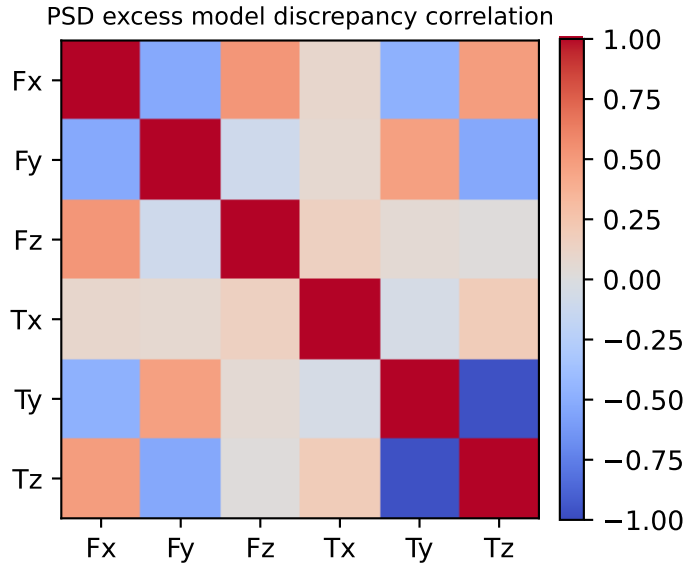
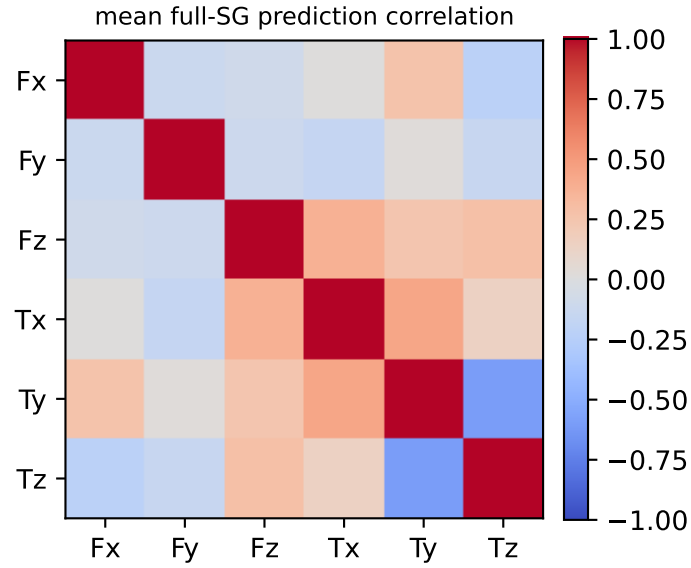
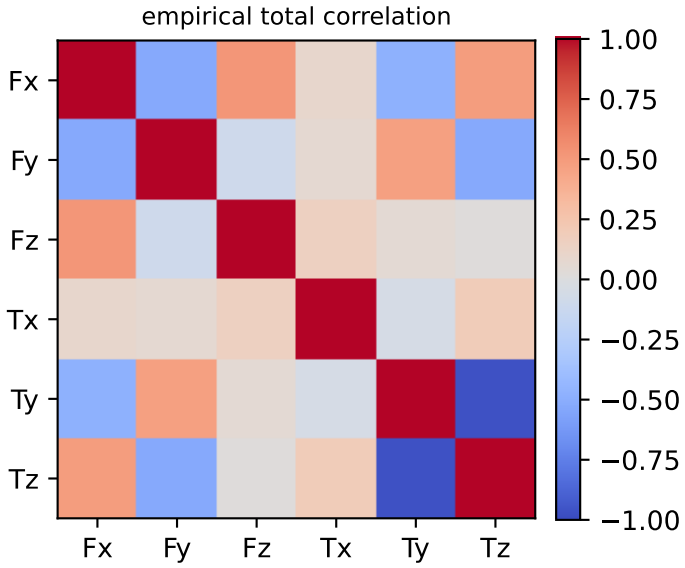
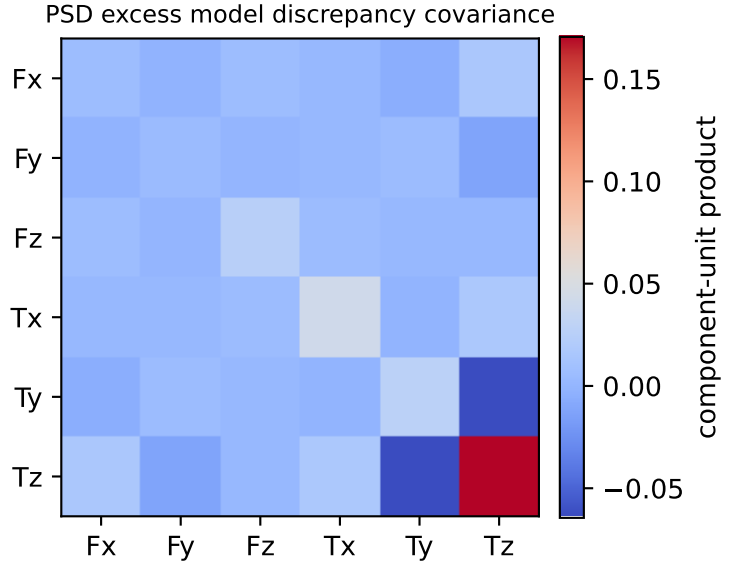
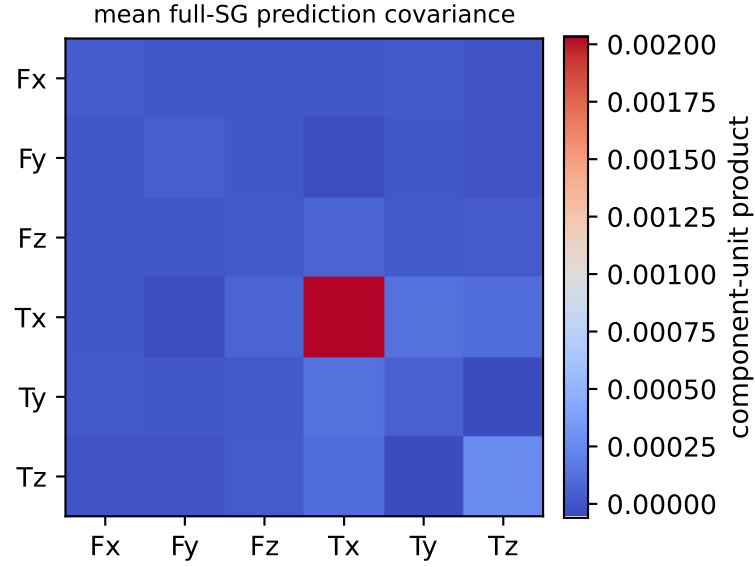
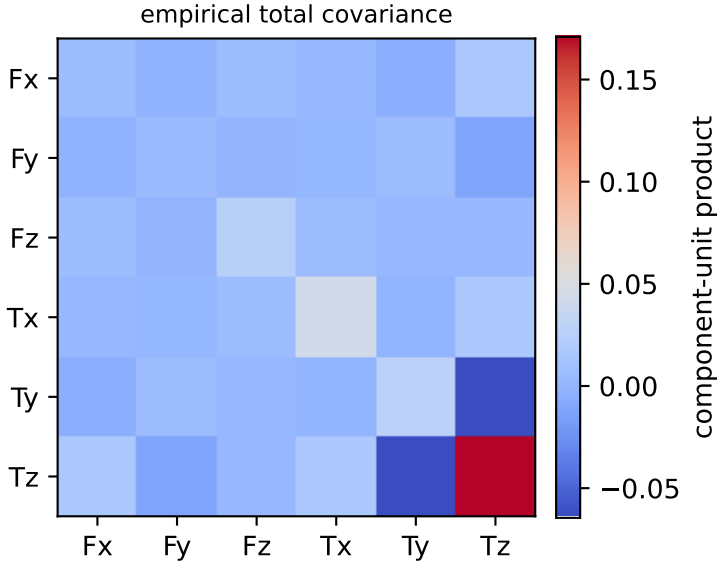


cog_y_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



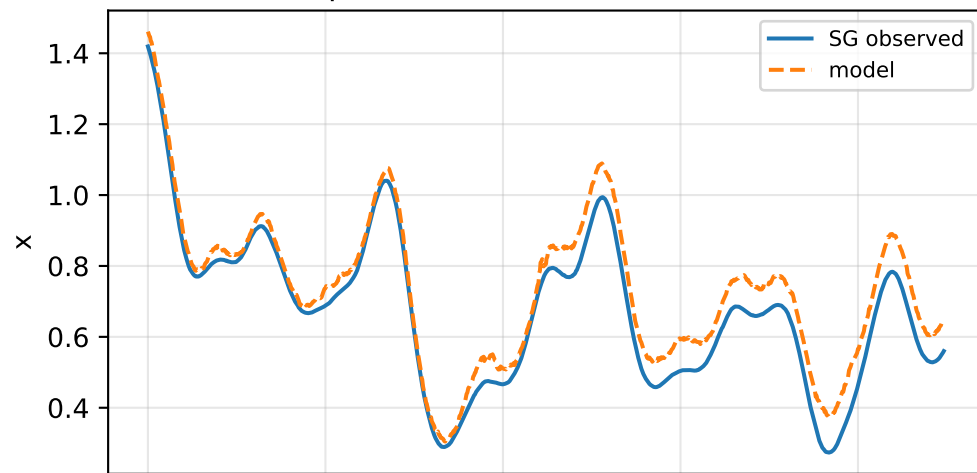
vector RMS about zero: force 0.23037 N, torque 0.4976 Nm; component std about mean: force [0.074 0.0614 0.1555], torque [0.2091 0.161 0.4134]

cog_y_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0. 0.002 0.003 0.025 0.042 0.199]

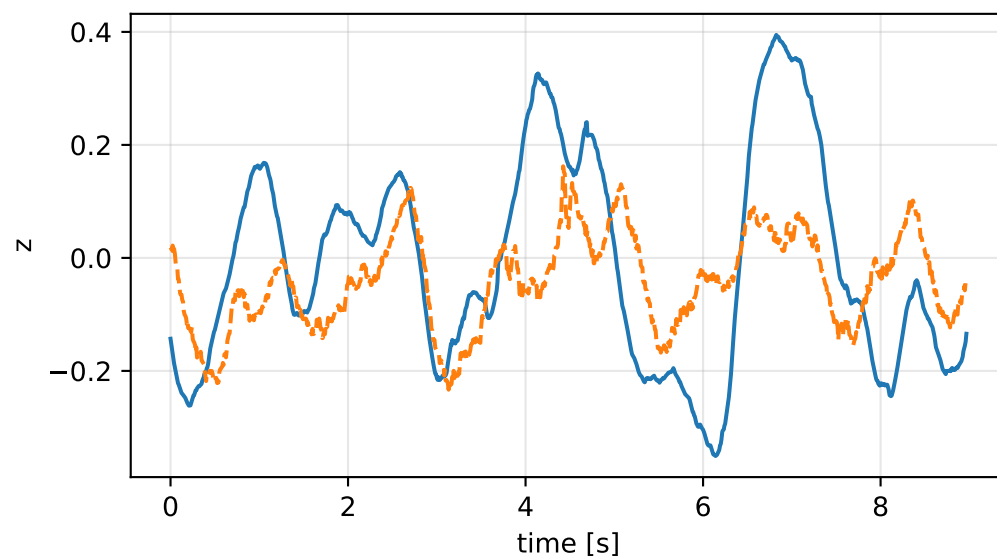
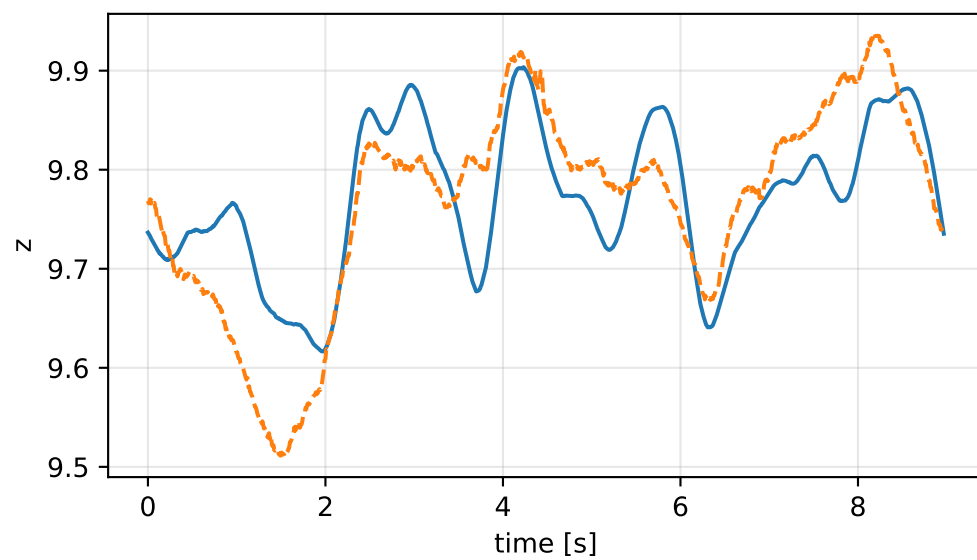
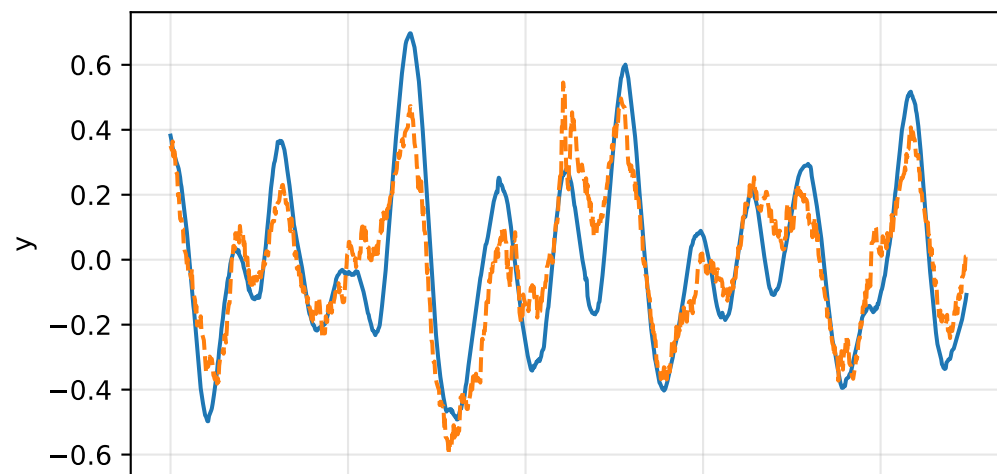
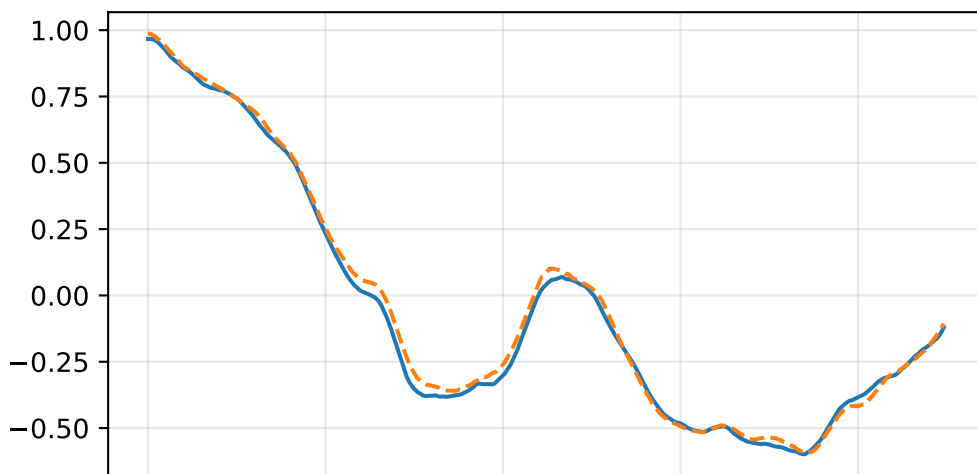
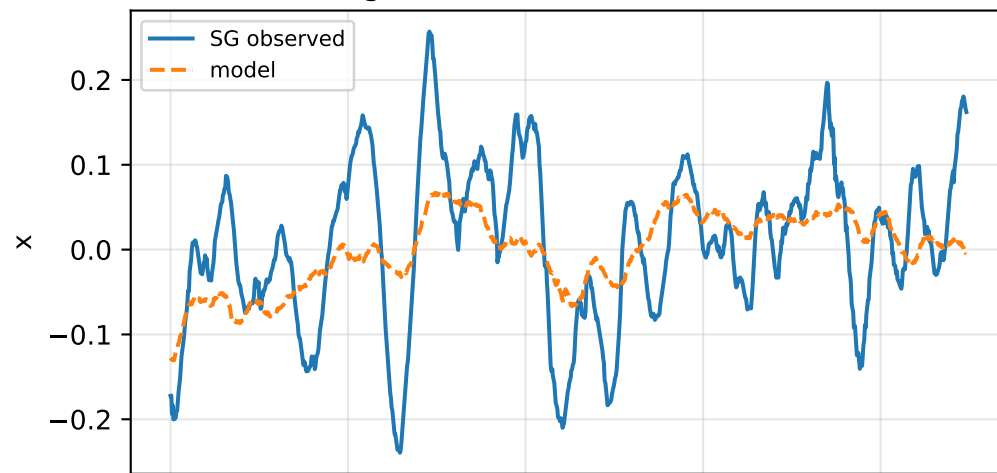


cog_y_nominal: acceleration residual objective

specific acceleration [m/s^2]

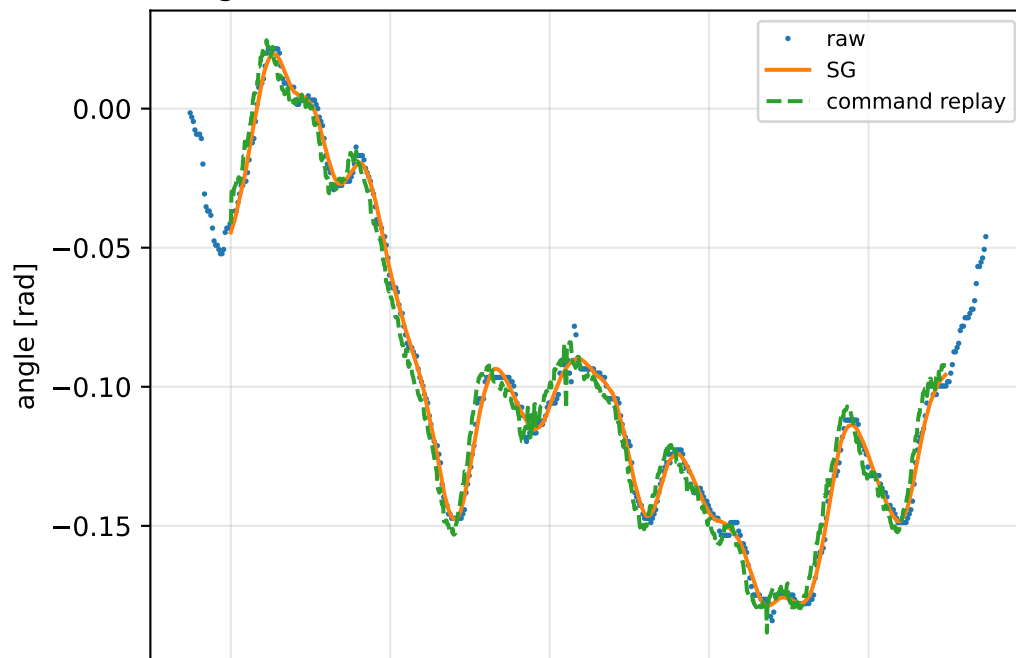


angular acceleration [rad/s^2]

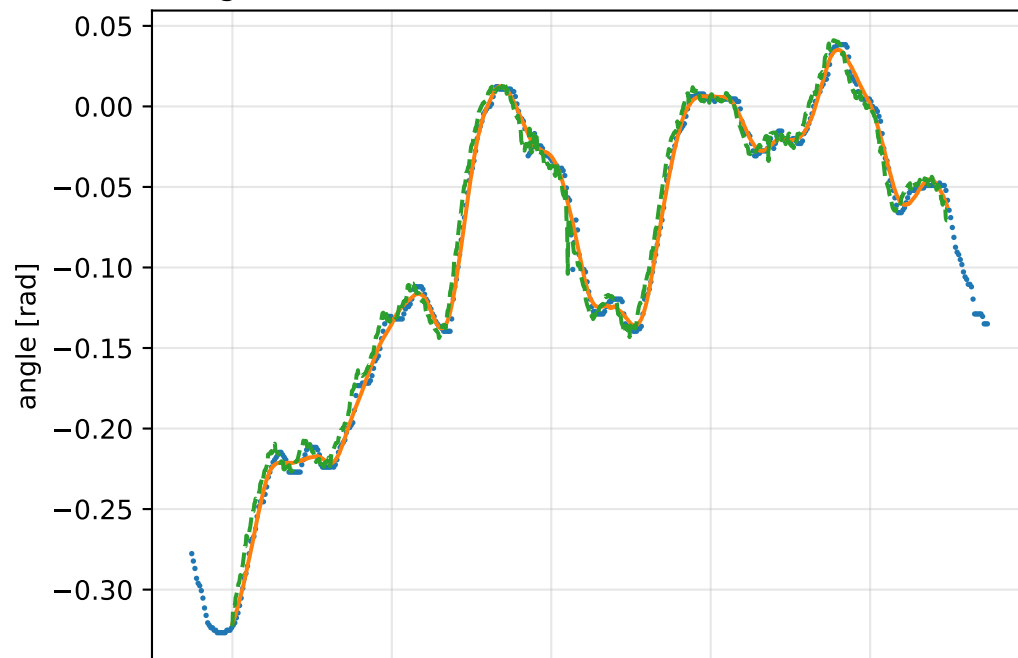


cog_y_nominal: gimbal measurement audit (objective=measured_sg)

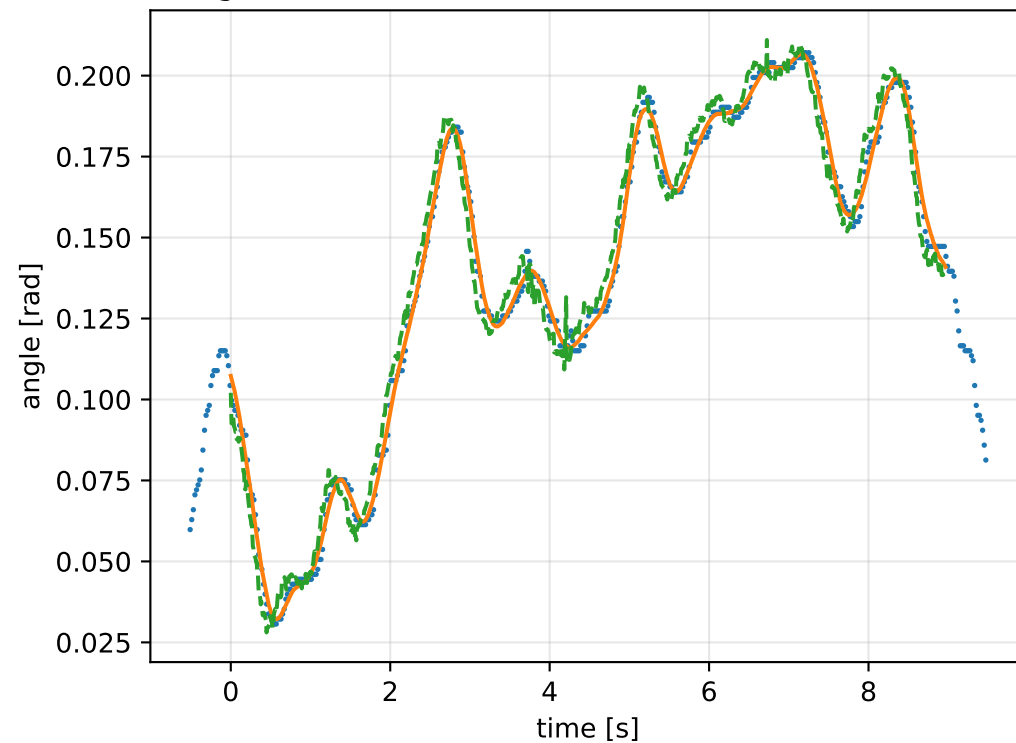
gimbal 1: RMSE=0.006345, max=0.01494 rad



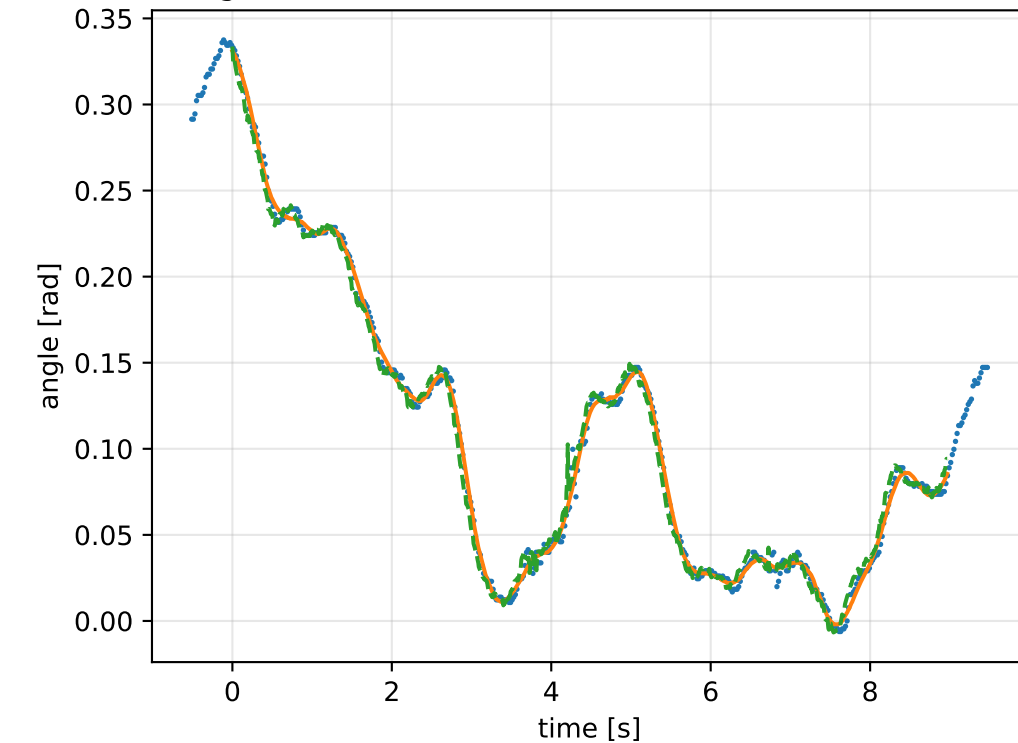
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

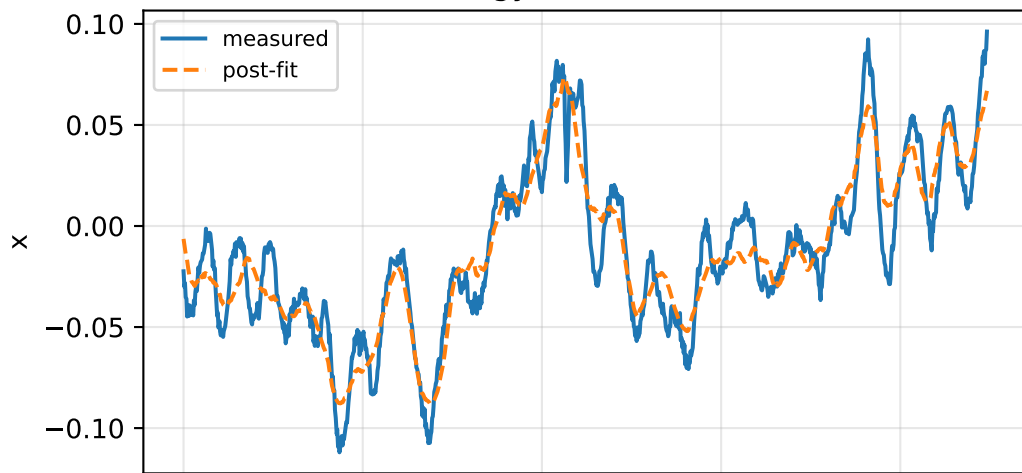


gimbal 4: RMSE=0.006321, max=0.03642 rad

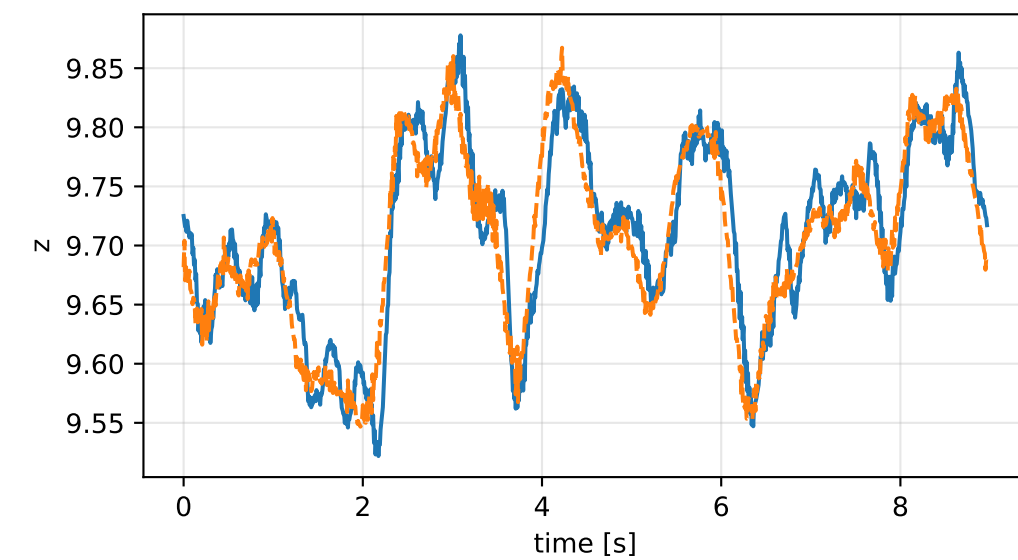
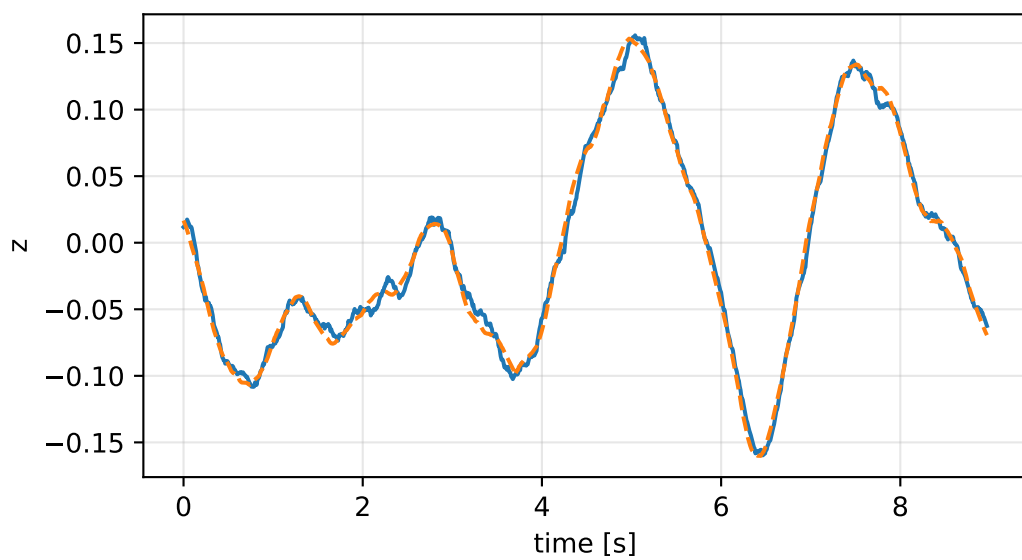
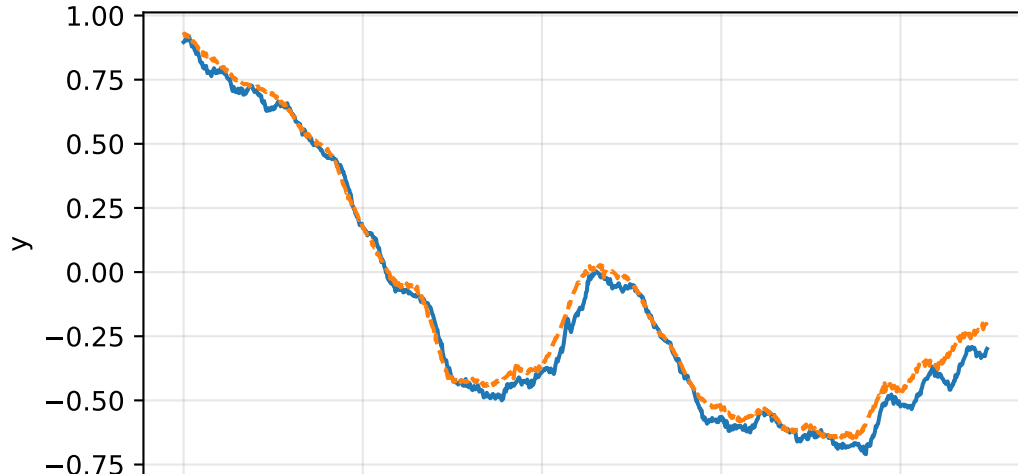
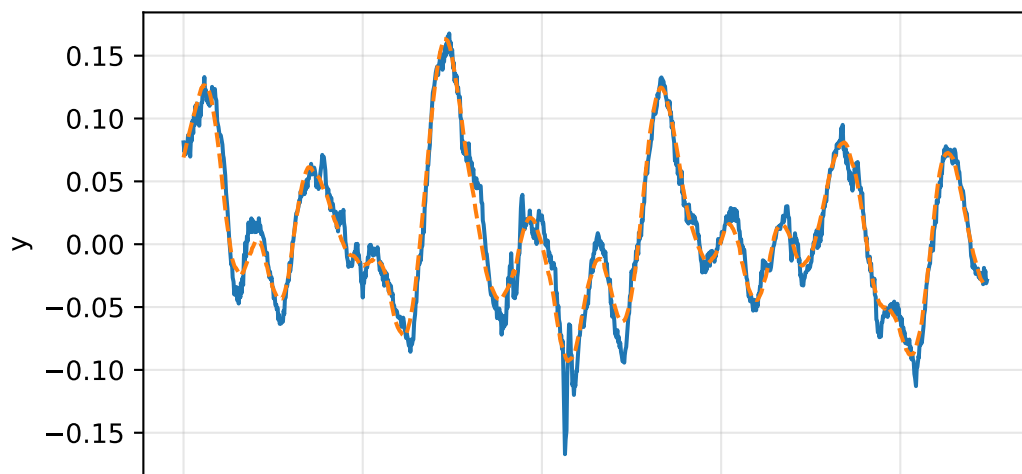
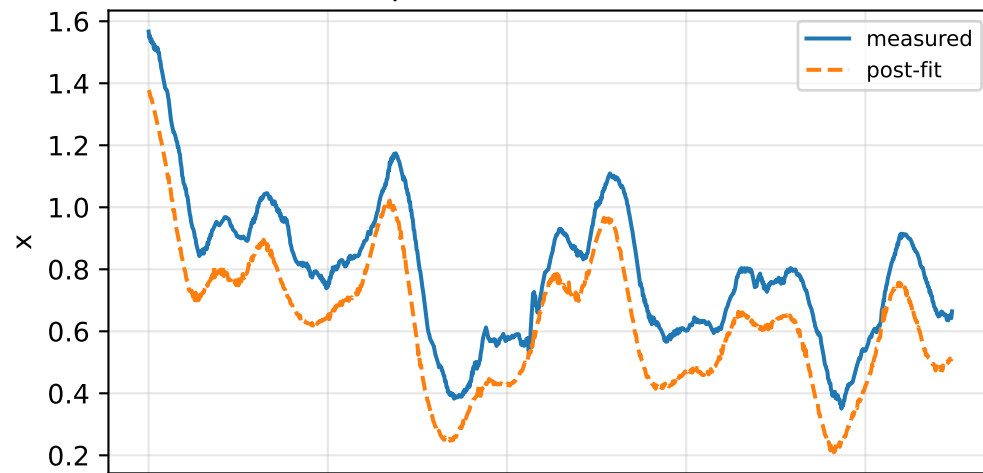


cog_y_nominal: IMU comparison (diagnostic only)

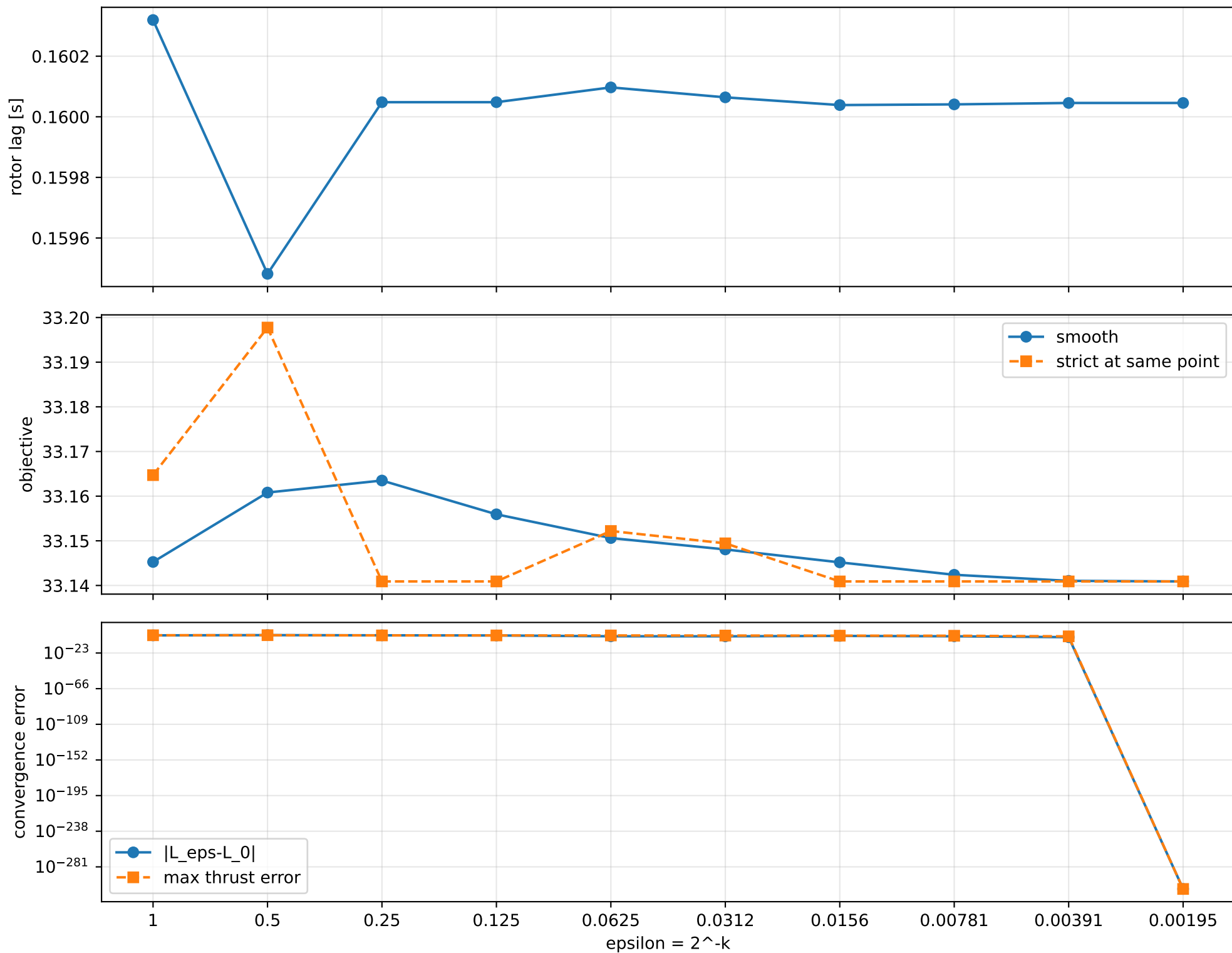
gyro [rad/s]



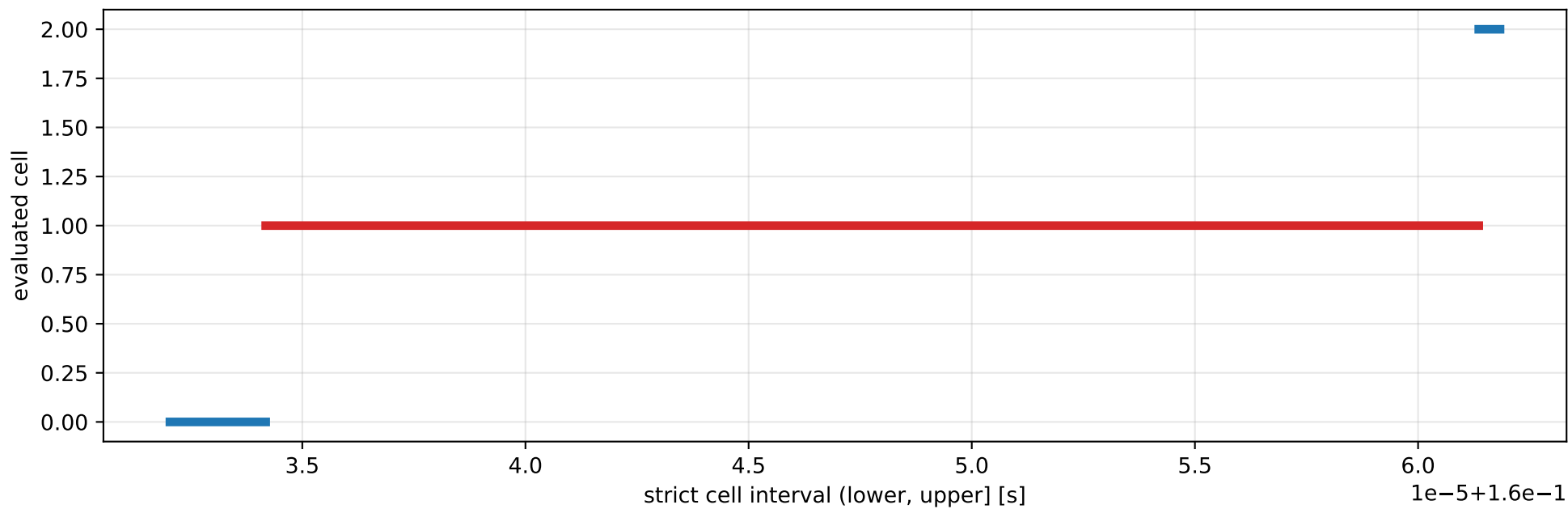
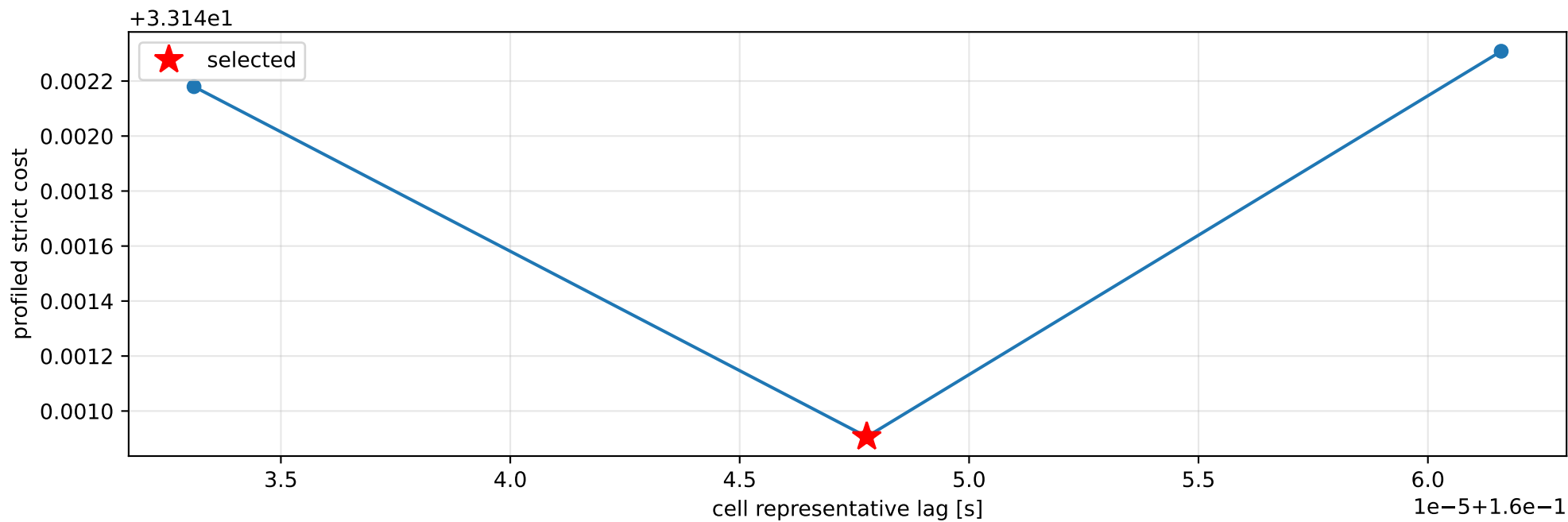
specific force [m/s^2]



cog_y_nominal: smooth-to-strict continuation



cog_y_nominal: exact strict-ZOH lag cells



selected (0.16003418, 0.160061359] s; final smooth 0.160045623 s; data boundary=False

cog_y_nominal: parameter estimate

Case: cog_y_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.98525302

inertia [kg m²]:

[[2.13076942 0.06818154 0.14453767]

[0.06818154 0.47322554 -0.64104982]

[0.14453767 -0.64104982 1.92363555]]

CoG body [m]: [-6.71894931e-02 -3.05313657e-05 -6.45833693e-02]

force effectiveness: [0.84899151 0.54166694 0.4841731 0.83970649]

scale-free J/m [m²]:

[[1.07329866 0.03434401 0.07280567]

[0.03434401 0.23837039 -0.32290585]

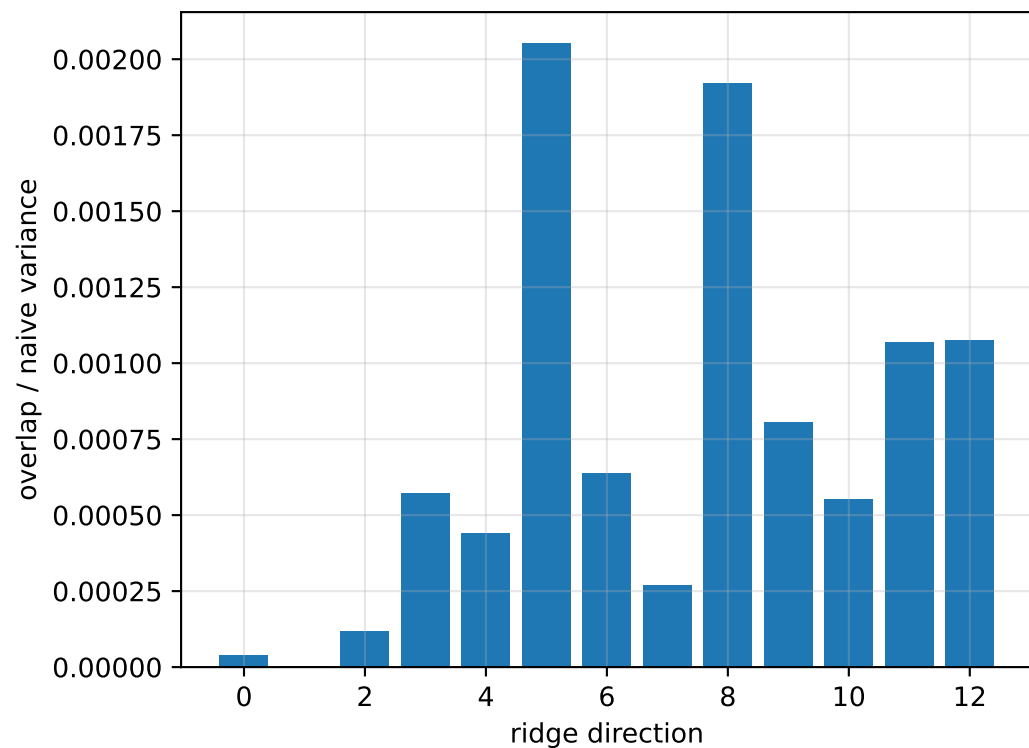
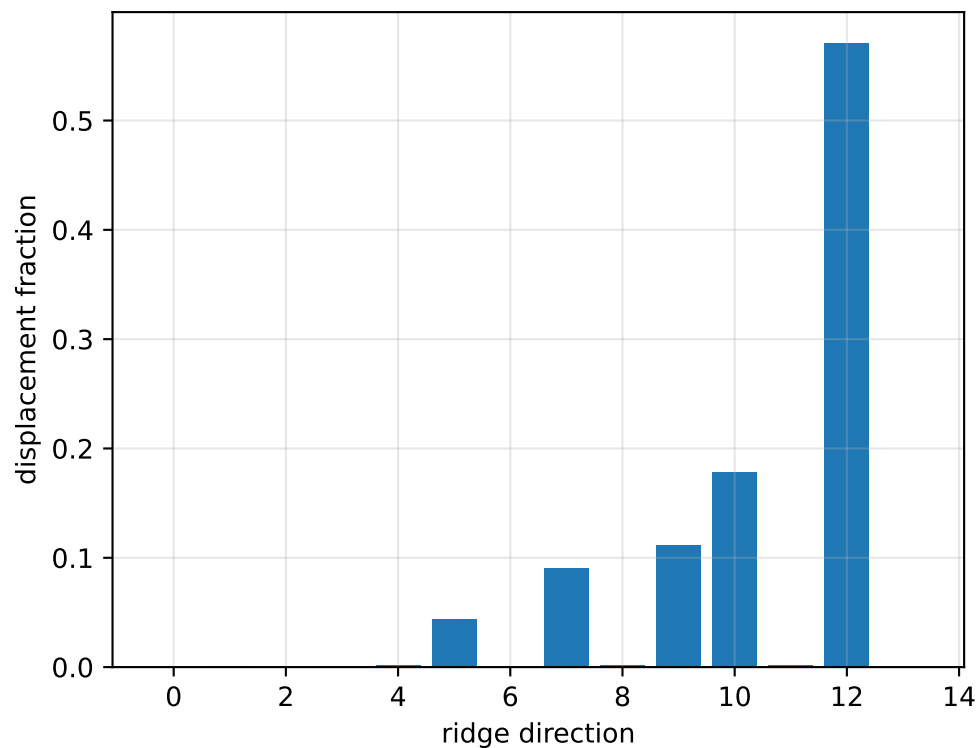
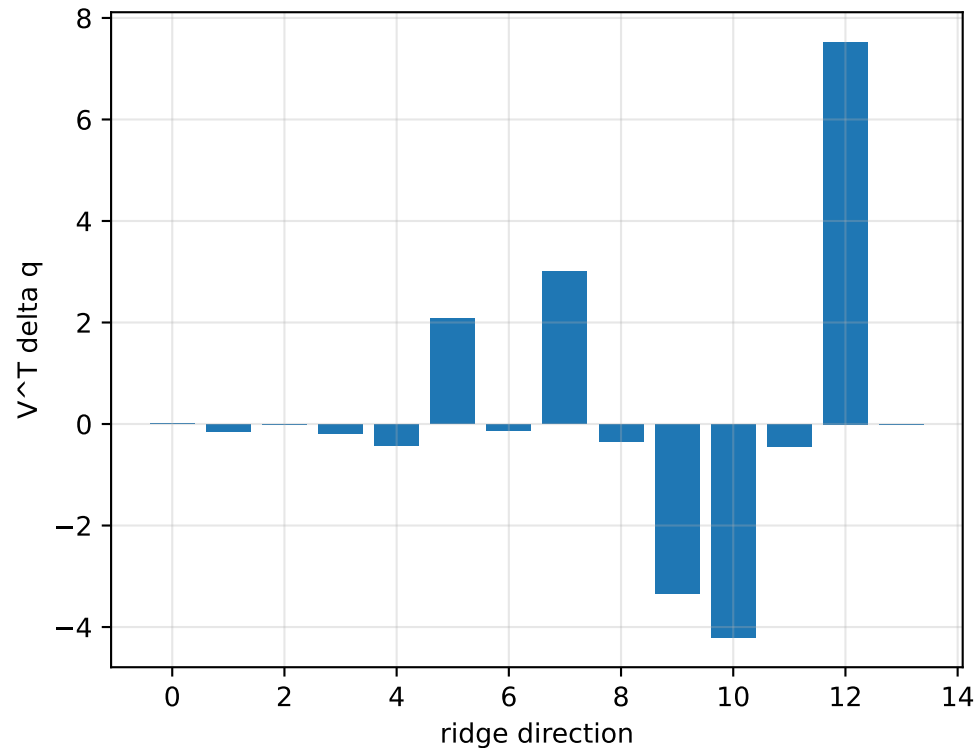
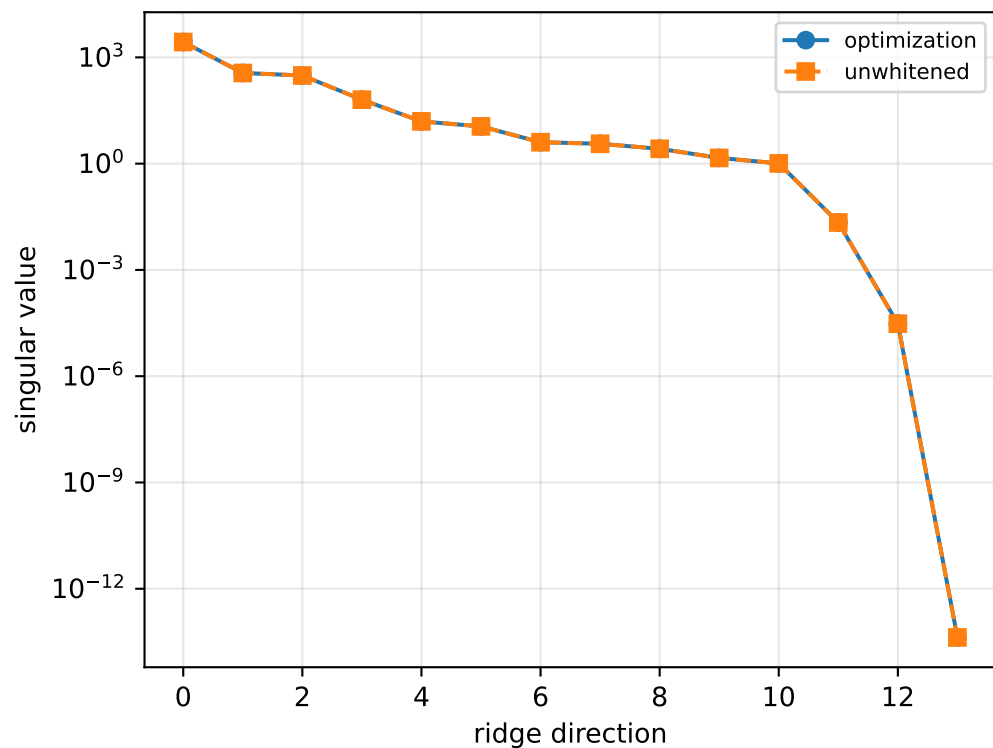
[0.07280567 -0.32290585 0.96896241]]

scale-free f/m: [0.42764902 0.27284529 0.24388483 0.42297202]

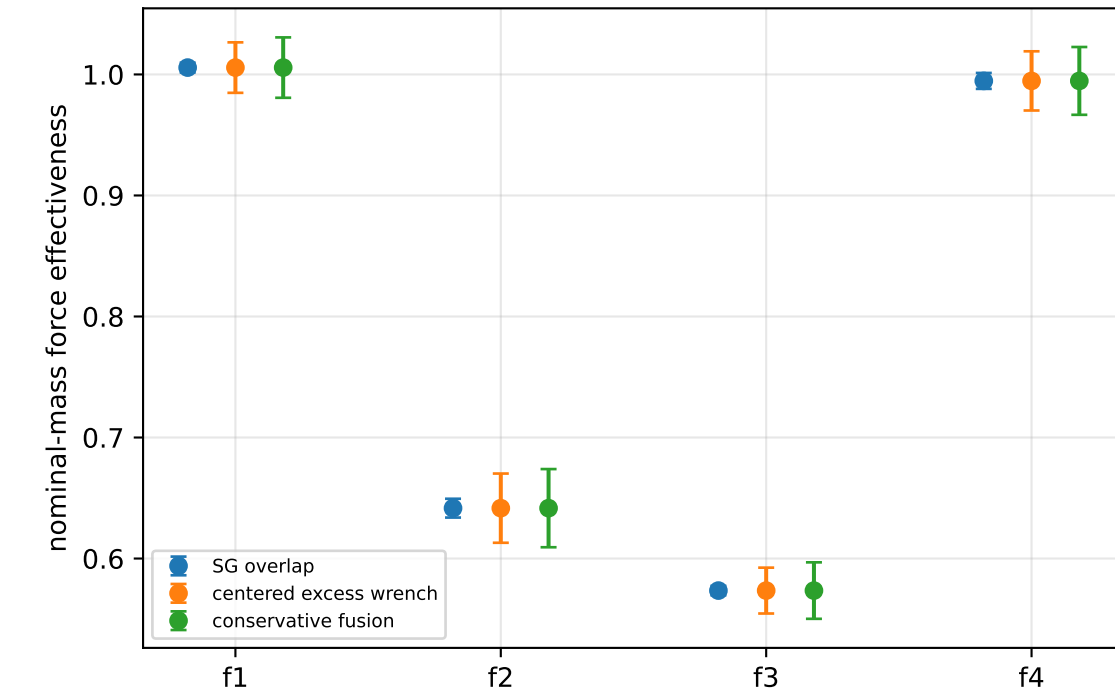
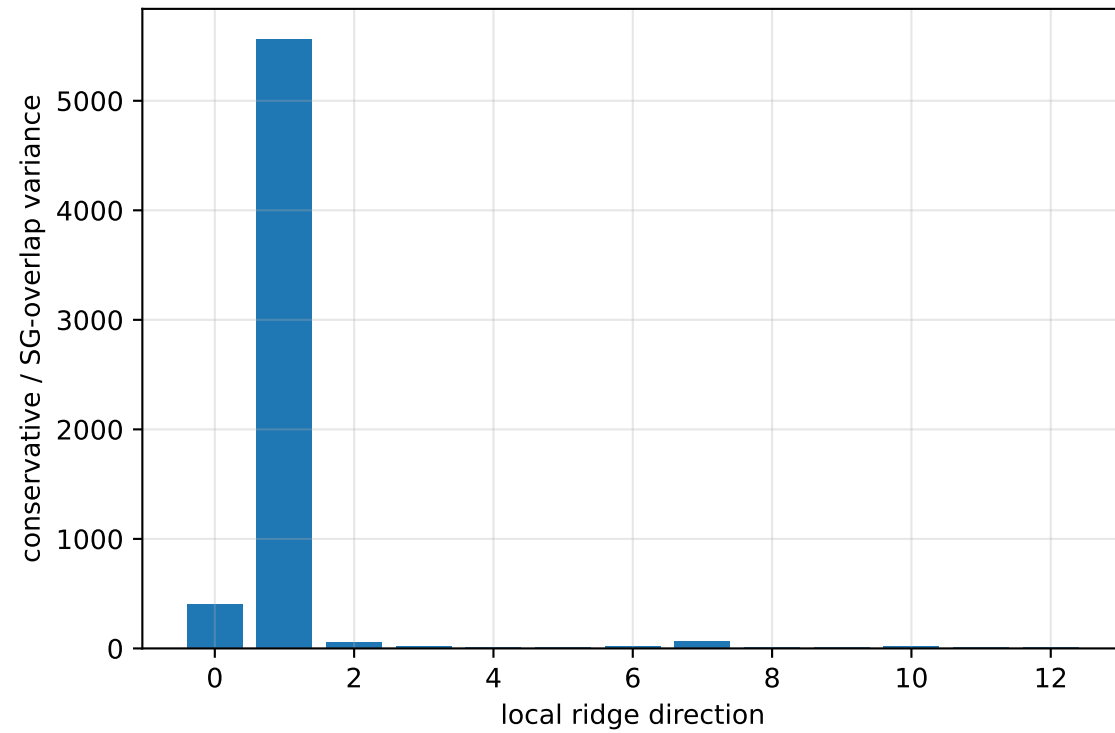
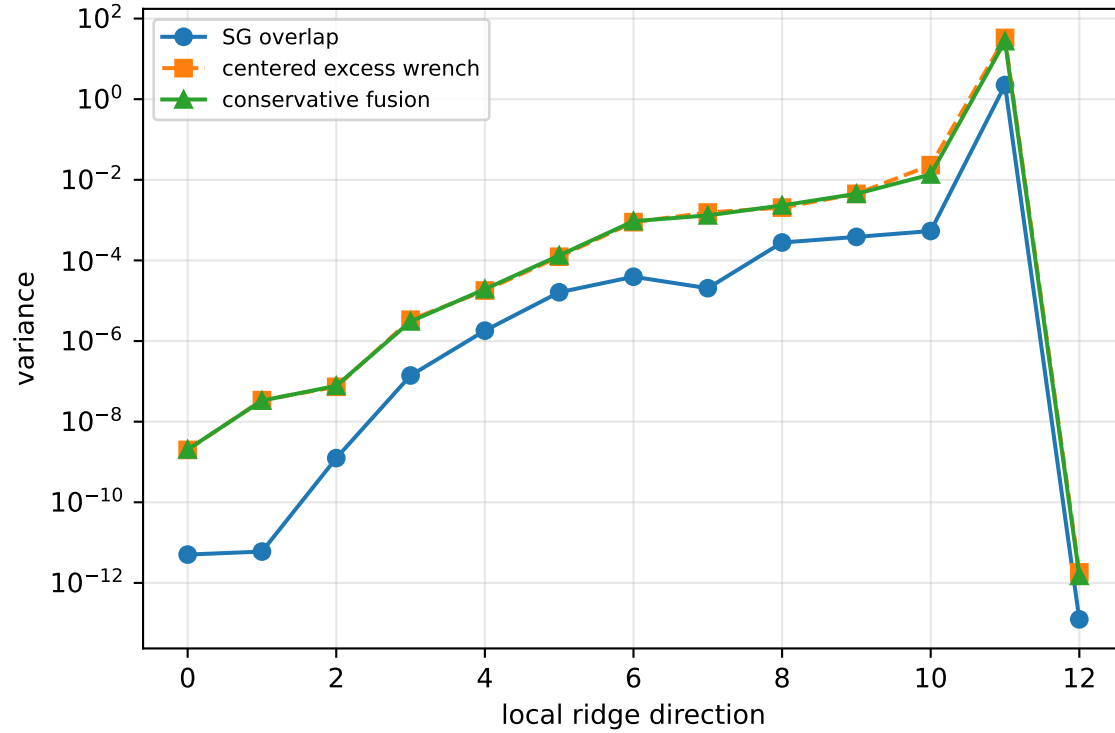
rotor lag cell: (0.16003418, 0.160061359] s

representative: 0.16004777 s

cog_y_nominal: ridge spectrum (rank 13, nullity 1, ||jv||=6.86e-14)



cog_y_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 387.7$
wrench-to-acceleration recovery max error = $3.09\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

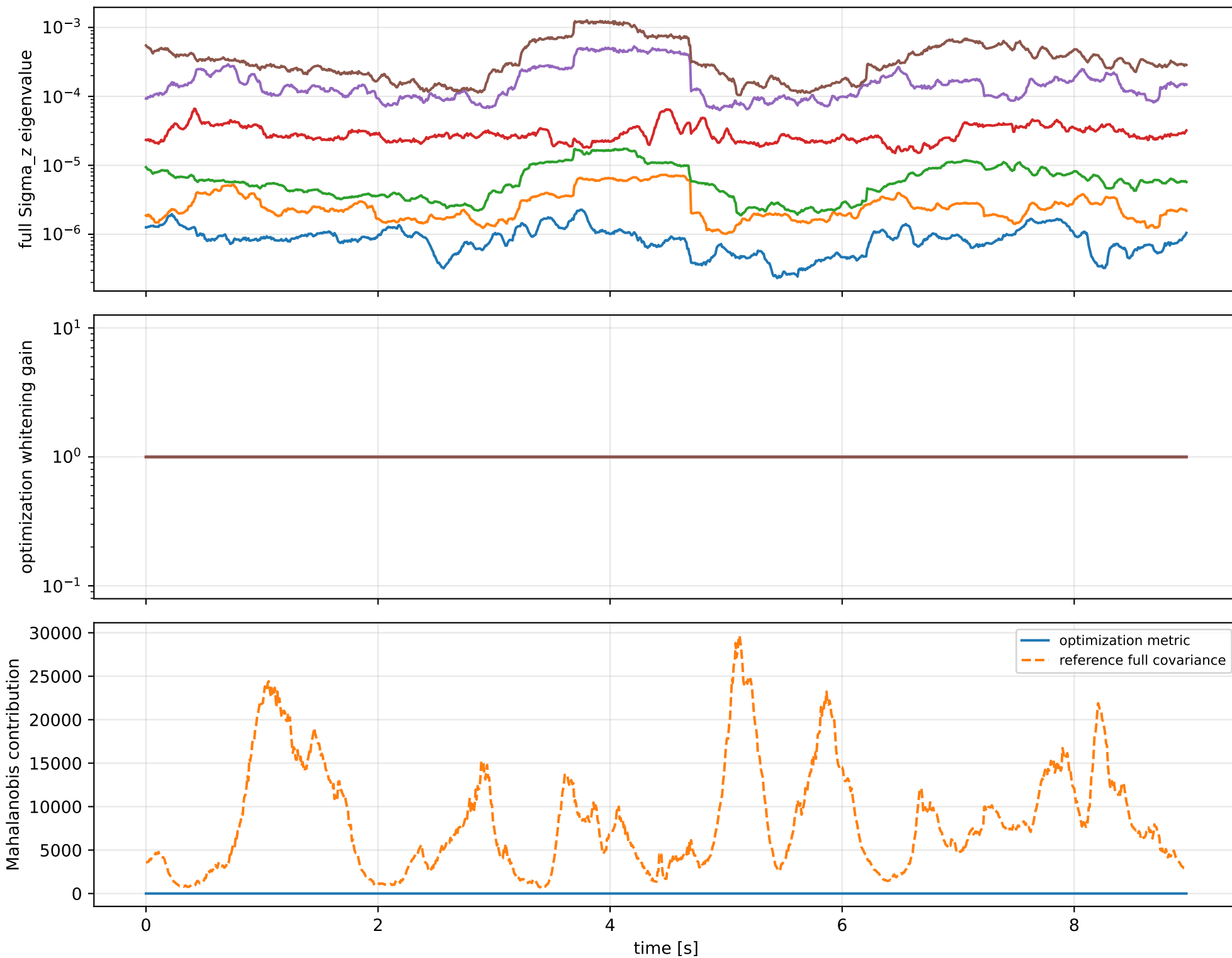
index 11: variance=27.52, ratio=12.26
 $\log_{\text{mass}}=+0.0243$; second_moment_diag_1=+0.438
second_moment_diag_2=-0.492; second_moment_diag_3=-0.0669
second_moment_offdiag_12=+0.696; second_moment_offdiag_13=-0.156
second_moment_offdiag_23=+0.221; cog_x=-0.000156
cog_y=+0.000511; cog_z=-0.000182
log_force_eff_1=+0.0229; log_force_eff_2=+0.0203
log_force_eff_3=+0.0266; log_force_eff_4=+0.0267

index 10: variance=0.01354, ratio=25.26
 $\log_{\text{mass}}=+0.116$; second_moment_diag_1=+0.00425
second_moment_diag_2=+0.213; second_moment_diag_3=-0.748
second_moment_offdiag_12=-0.108; second_moment_offdiag_13=-0.533
second_moment_offdiag_23=+0.147; cog_x=-0.00717
cog_y=-0.0122; cog_z=+0.0179
log_force_eff_1=+0.189; log_force_eff_2=+0.136
log_force_eff_3=-0.00721; log_force_eff_4=+0.0973

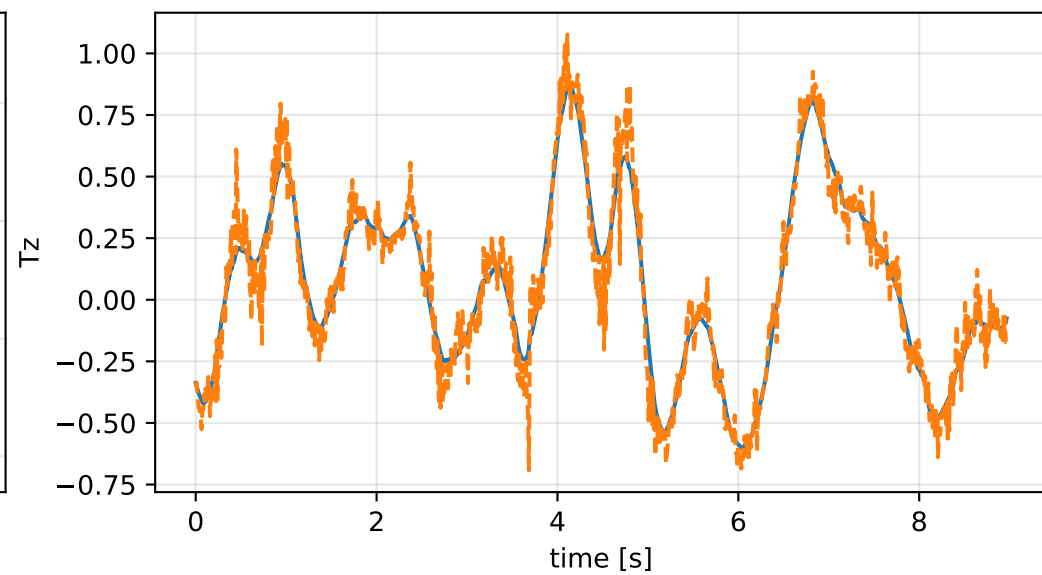
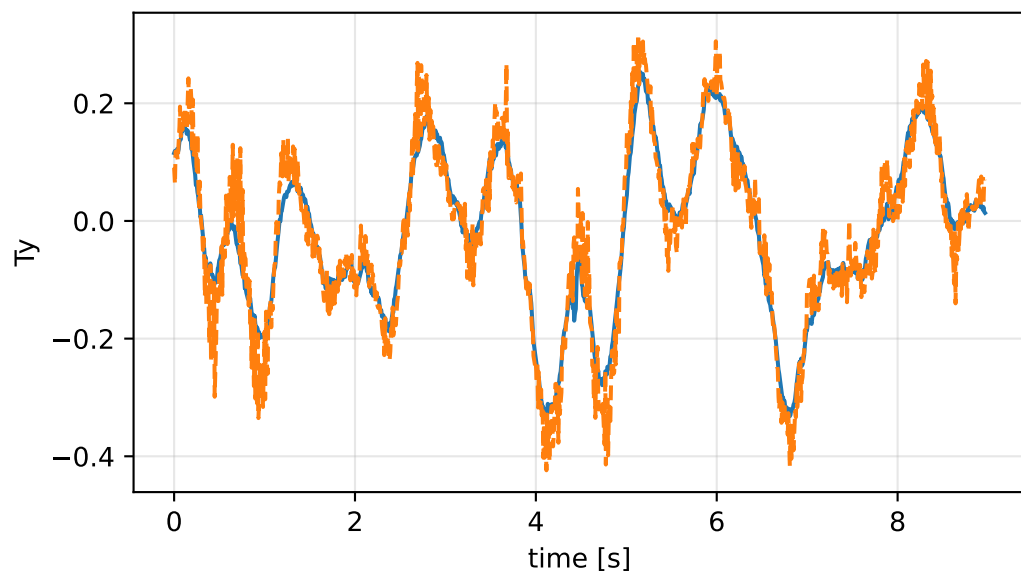
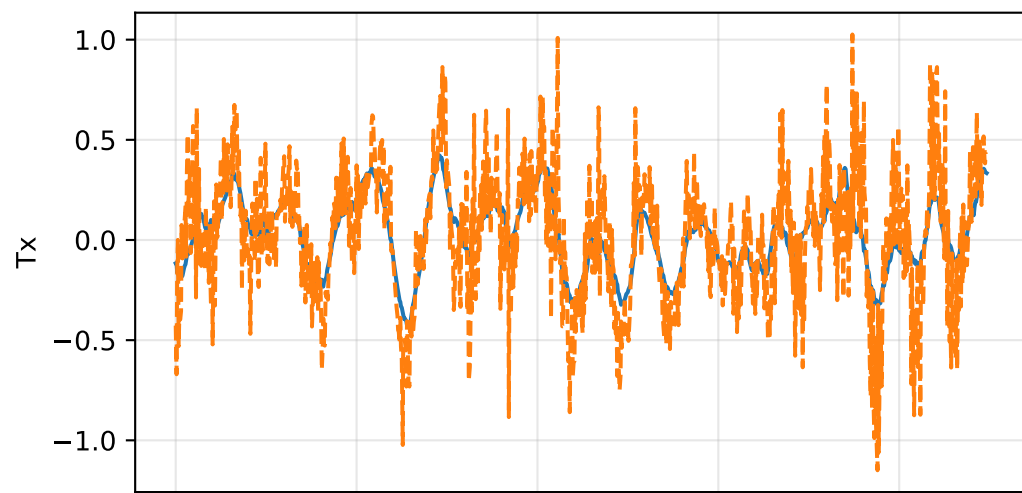
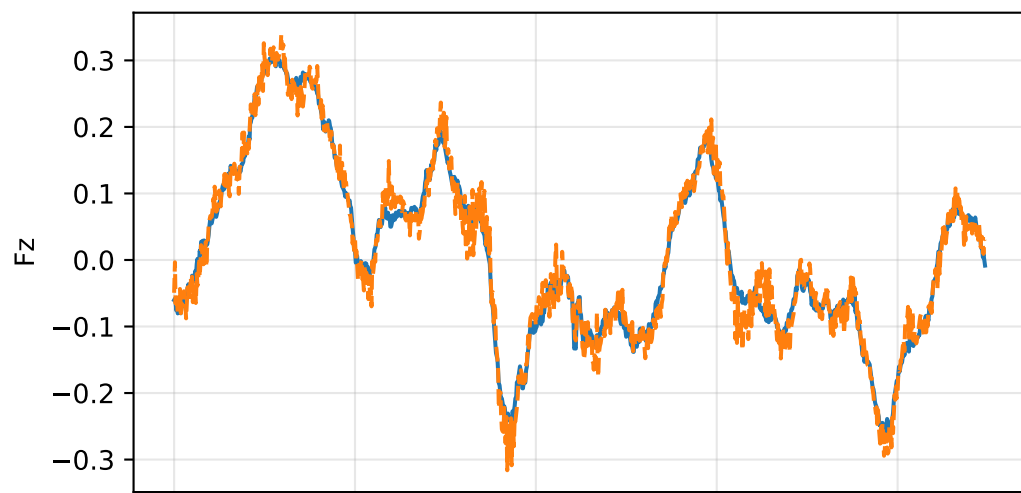
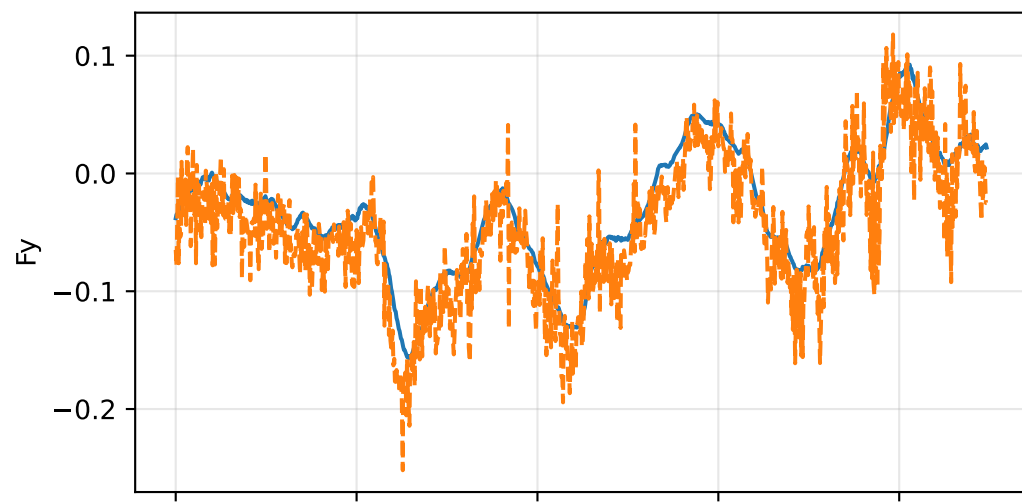
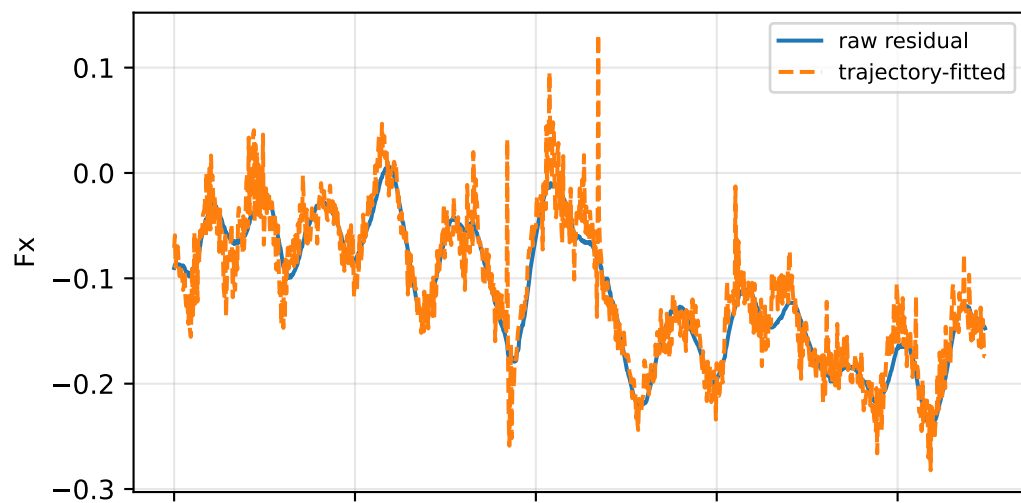
index 9: variance=0.004488, ratio=11.71
 $\log_{\text{mass}}=+0.0614$; second_moment_diag_1=-0.008
second_moment_diag_2=-0.215; second_moment_diag_3=-0.284
second_moment_offdiag_12=+0.0607; second_moment_offdiag_13=+0.311
second_moment_offdiag_23=-0.561; cog_x=+0.0396
cog_y=-0.0521; cog_z=+0.0312
log_force_eff_1=+0.108; log_force_eff_2=+0.613
log_force_eff_3=-0.0304; log_force_eff_4=-0.246

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog_y_nominal: optimization vs reference covariance



cog_y_nominal: wrench / closure (force max $6.15\text{e-}15$, torque max $2.22\text{e-}16$)



cog_y_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_cog_y_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/cog_y_nominal.j

SHA256: 3efc6d36ba4fff0dc12548d07dd00c0a5c4bcb02e99decbl3ffa2adae4ce3f12

data objective: 33.1409062722

prior objective: 1.14565044256e-07

total objective: 33.1409063867

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: cog_y_nominal (cog_position_body_m)

components: y

target: [-3.05265789e-05]

std: [1.e-05]

final: [-3.05313657e-05]

error: [-4.78675348e-09]

standardized residual: [-0.00047868]

factor objective: 1.14565044256e-07